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Roll No.....



ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.

(An ISO 9001:2015 Certified College)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

B. TECH DEGREE FIRST INTERNAL TEST- NOVEMBER 2022

FIFTH SEMESTER

CST 393 CRYPTOGRAPHIC ALGORITHMS (Honours)

Time: 2 Hours

Max.Marks: 50

PART A

Answer all Questions (3x5=15 marks)

Qn. No.	Questions	Marks	K level	CO & KL
✓ 1	Differentiate passive attack from active attack with example.	3	K1	CO1&K3
2	Define Transposition cipher. Convert the Given Text "CRYPTOGRAPHY" into cipher text using Rail fence Technique.	3	K2	CO1&K3
3	Brief the strengths of triple DES.	3	K2	CO2&K3
4	Compare a block cipher and a stream cipher?	3	K1	CO2&K3
✓ 5	List the principle elements of a Public-key Cryptosystem.	3	K1	CO3&K3

PART B

Answer five questions (7 x 5 = 35 Marks)

Module 1: Qn. No 6,7 and 8 – Answer any 2

Module 2: Qn. No 9,10 and 11 – Answer any 2

Module 3: Qn. No 12 is mandatory

Qn. No.	Questions	Marks	K Level	CO & KL
✓ 6	(a) What are the 3 aspects of security?	3	K1	CO1 & K3
	(b) Describe the OSI Security architectural model using a diagram.	4	K1	CO1 & K3
7	Discuss about different polyalphabetic cipher. <i>playfair, hill, vigenere</i>	7	K2	CO1 & K3
✓ 8	Encrypt the text "PAYMOREMONEY" using the Hill cipher with the key $\begin{pmatrix} 17 & 17 & 5 \\ 21 & 18 & 21 \\ 2 & 2 & 19 \end{pmatrix}$ Show the calculations.	7	K3	CO1 & K3
9	(a) What is avalanche and completeness effect in DES?	3	K1	CO2 & K3
	(b) Elucidate the confusion and diffusion properties of modern block ciphers.	4	K2	CO1 & K3

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10	Explain the key generation in IDEA.	7	K2	CO2 & K3
11	Describe the AES algorithm with suitable block diagram.	7	K2	CO2 & K3
12	Draw the general structure of DES and explain the encryption decryption process.	7	K2	CO2 & K3

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ST. JOSEPH'S COLLEGE OF ENGINEERING AND TECHNOLOGY, PALAI.*(An ISO 9001:2015 Certified College)***DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
B. TECH DEGREE SECOND INTERNAL TEST- DECEMBER 2022****FIFTH SEMESTER – (HONOURS)
CST 393 CRYPTOGRAPHIC ALGORITHMS**

Time: 2 Hours

Max.Marks: 50

PART A

Answer all Questions (3x5=15 marks)

Qn. No.	Questions	Marks	K level	CO & KL
1	State the possible attacks in RSA.	3	K1	CO3&K3
2	Briefly explain the idea behind the knapsack cryptosystem.	3	K2	CO3&K3
3	Describe the following (i) Updating keys (ii) Compromised Keys.	3	K2	CO4&K3
4	What is the role of Key Distribution centre?	3	K1	CO4&K3
5	Define Message Authentication?	3	K1	CO5&K3

PART B

Answer five questions (7 x 5 = 35 Marks)

Module 3: Qn. No 6,7 and 8 – Answer any 2

Module 4: Qn. No 9,10 and 11 – Answer any 2

Module 5: Qn. No 12 is mandatory

Qn. No.	Questions	Marks	K Level	CO & KL
6	Illustrate ElGamal encryption and decryption algorithm.	7	K2	CO3&K3
7	Demonstrate Diffie Hellman Key exchange algorithm.	7	K2	CO3&K3
8	Perform encryption using RSA Algorithm for the following $p=7$; $q=11$; $e=13$; $M=5$.	7	K3	CO3&K3
9	List four general categories of schemes for the distribution of public keys	7	K2	CO4&K3
10	Explain the Symmetric key distribution using asymmetric encryption.	7	K2	CO4&K3
11	List the duties and core components of a PKI? Briefly describe each component.	7	K1	CO4&K3
12	(a) Define the classes of message authentication function.	4	K2	CO5&K3
	(b) What are the security requirements of message authentication?	3	K1	CO5&K3

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech (Hons.) Degree Examination December 2022 (2020 Admn.)

Course Code: CST 393**Course Name: CRYPTOGRAPHIC ALGORITHMS**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

- | | | |
|-----|---|---|
| ✓ 1 | Comment on Passive and active attacks on the encryption schemes. | 3 |
| ✓ 2 | Write the rules of play fair ciphering technique. | 3 |
| ✓ 3 | Explain the generation of round keys in DES. | 3 |
| ✓ 4 | Brief the construction of S-box. | 3 |
| ✓ 5 | Mention the three uses of an encryption scheme. | 3 |
| 6 | Define one-way function and trap-door one way function. | 3 |
| 7 | State a method to identify the errors that may happen during transmission of key values. | 3 |
| ✓ 8 | If the key values are compromised, what are the steps to be taken by Alice to prevent further issues? | 3 |
| 9 | Define message digest. | 3 |
| 10 | Draw a neat a sketch of using MAC for authentication. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

- | | | |
|----|--|----|
| 11 | a) Draw the basic model of network security and explain each term. | 10 |
| | b) Compare stream cipher and block cipher with example. | 4 |

OR

- | | | |
|----|--|---|
| 12 | a) Encrypt the text " <i>I only regret that I have but one life to give for my country</i> " using transposition cipher with the key (3,2,1,4,5). Show decryption of the ciphertext to recover the original text back. | 6 |
| | b) Encrypt the message " <i>the house is being sold tonight</i> " using the following ciphers. Ignore the space between words. | 8 |
| | i. Vigenere cipher with key = " <i>largest</i> ". | |
| | ii. Autokey system of Vigenere cipher with key = " <i>largest</i> ". | |

Module -2

- 13 a) Explain the encryption and decryption of triple DES using 2 keys and 3 keys. 5
 b) Summarize the primitive operations in RC4 algorithm. 9

OR

- 14 a) Illustrate Linear and differential cryptanalysis. 10
 b) Sketch the diagram for Fiestel structure. 4

Module -3

- 15 a) Discuss Elliptic curve cryptography. 10
 b) Write the equation for the addition of two points on the elliptic curve. 4

OR

- 16 a) Explain Diffie Hellman Key Exchange. 8
 User A and B use the Diffie-Hellman key exchange technique with a common prime $q=17$ and primitive root $\alpha=7$. If user A has private key $X_A=3$, and user B has private key $X_B=6$, what is the secret key shared?
 b) How does a man-in-the-middle attack happen to DH key exchange? 6

Module -4

- 17 a) Explain the different PKIX management protocols. 6
 b) Explain the concept of symmetric key distribution using asymmetric keys. 8

OR

- 18 With neat diagram, explain Public Key Infrastructure (PKI). 14

Module -5

- 19 a) Illustrate Message Authentication Code (MAC) and HMAC. 8
 b) Specify the format for X.509 certificate. Explain the steps required to obtain user's certificate. 6

OR

- 20 Explain SHA-512 algorithm. 14
